INCH-POUND

MIL-PRF-27/303C
18 July 2008
SUPERSEDING
MIL-T-27/303B
2 June 2003

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER, AND HIGH-POWER PULSE)
INDUCTOR, POWER

Inactive for new design
after 13 March 1991

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

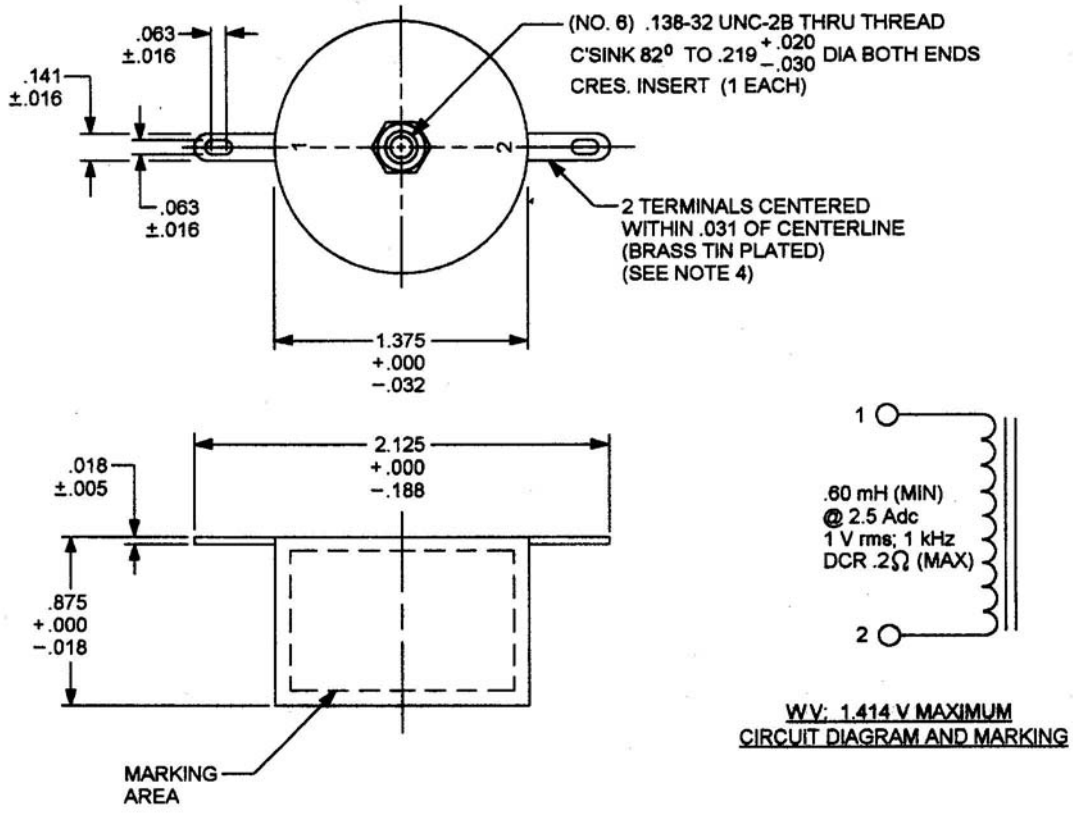


FIGURE 1. Dimensions and configurations.

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INCHES	MM	INCHES	MM
.005	0.13	.138	3.51
.016	0.41	.141	3.88
.018	0.46	.188	4.78
.020	0.51	.219	5.56
.030	0.76	.875	22.22
.032	0.81	2.135	53.98
.063	1.60		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.
4. Terminals shall be brass, 70 percent Cu, 30 percent Zn in accordance with ASTM-B36/B36M and ASTM-B121/B121M

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Inductance (1-2): 0.60 millihenry, minimum.

Voltage (1-2): 1.0 volts rms, 1 kHz.

Current (1-2): 2.5 amperes dc.

Working voltage (1-2): 1.414 volts, maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder type, brass, tin plated.

Length: .391 inch, maximum.

Weight: 85.05 grams, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 100 volts rms.

Electrical characteristics:

Inductance: With 1.0 volt rms, 1 kHz, and 2.5 amperes dc applied to (1-2); the inductance shall be 0.60 millihenry, minimum.

DC resistance (1-2): 0.2 ohm, maximum.

Temperature rise: 50°C with 1.0 volt rms, 1 kHz, 2.5 amperes dc applied to (1-2) at an ambient temperature of 80 °C, maximum.

Vibration (high frequency): MIL-STD-202, method 204, test condition A.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

VERIFICATION:

Qualification inspection: Not applicable.

Quality conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

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Marking location: See figure 1.

Part or Identification Number (PIN): M27/303-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

ASTM B36/36M
ASTM B121/B121M
MIL-PRF-27
MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2008-037)

Review activities:
Army - AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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